

Convolution and Multiplication Semantics on Signed Hypergraphs

The algebraic foundation for HyMeKo's `aggregate` extension

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Abstract

HyMeKo already represents structure, observation, and reward as one signed (weighted) hypergraph. The *computation* over it — message passing, balance, fuzzy firing, reward — is, in every case, a **semiring-valued convolution of the signed incidence with a vertex field**. This note gives the algebraic object precisely (a commutative-monoid aggregation, a semiring for the signed-weighted product), the conditions under which it is well defined, the catalog of admissible algebras (the `aggregate` tags), and the differential/spectral structure each one induces. It is the foundation on which the `aggregate` language extension (docs/plans/2026-06-25-hymeko-aggregation-semantics) can be built soundly.

1 Setup: signed hypergraph and its boundary operator

A *signed weighted hypergraph* is $H = (V, E, w, \sigma)$ with vertices V ($n = |V|$), hyperedges $E \subseteq 2^V$ ($m = |E|$), arc weights $w_{ev} \in \mathbb{R}$ and signs $\sigma_{ev} \in \{+1, -1\}$ for each incident pair $v \in e$. Its *star-expansion* promotes each hyperedge e to a hub node joined to its members by signed, weighted arcs; the data is the *signed incidence*

$$B \in \mathbb{R}^{n \times m}, \quad B_{ve} = \sigma_{ev} w_{ev} \text{ if } v \in e, \text{ else } 0.$$

B is the discrete *boundary operator* $\partial : C_1 \rightarrow C_0$ of the chain complex (hyperedges are 1-chains, vertices 0-chains); $B^\top$ is the coboundary δ . A *vertex field* is $x : V \rightarrow S$ valued in an algebra S (a feature, a state, a membership). **Hyperedges may themselves be members of hyperedges** (the star-expansion is closed): incidence then composes, $B = B^{(2)} \dots B^{(1)}$ (the “hyperedge-on-hyperedge” P -space; §7).

2 The core object: semiring-valued convolution

Definition 1 (Aggregation algebra). *An aggregation algebra is a commutative monoid $(S, \oplus, \mathbf{0})$: $\oplus$ associative and commutative with identity $\mathbf{0}$. A signed-weighted aggregation additionally needs a second operation $\otimes$ (the arc weight/sign acting on a value) such that $(S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ is a semiring: $(S, \oplus, \mathbf{0})$ a commutative monoid, $(S, \otimes, \mathbf{1})$ a monoid, $\otimes$ distributes over $\oplus$, and $\mathbf{0}$ annihilates ($\mathbf{0} \otimes a = \mathbf{0}$).*

Definition 2 (Hypergraph convolution). *For a vertex field $x \in S^V$ the convolution onto the hyperedges is $(B \otimes x) \in S^E$ with*

$$(B \otimes x)_e = \bigoplus_{v \in e} (B_{ve} \otimes x_v) = \bigoplus_{v \in e} \sigma_{ev} w_{ev} \otimes x_v.$$

This is exactly the semiring matrix-vector product $B^\top \otimes x$. The dual convolution $B \otimes y$ (with $y \in S^E$) scatters edge values back onto vertices.

Proposition 1 (Well-definedness). $(B \otimes x)_e$ is independent of the order in which the members of e are visited iff $\oplus$ is commutative and associative; it is defined for the empty edge and for padded/masked members iff $\oplus$ has identity $\mathbf{0}$ (the pad value); it composes across nested hyperedges (§7) iff $\otimes$ distributes over $\oplus$. Hence: **the convolution is well defined precisely when the declared algebra is a commutative monoid, and signed-weighted convolution requires a semiring.**

Sketch. Members of a hyperedge are an unordered (multi)set, so order independence is exactly commutativity+associativity of $\oplus$; the identity supplies the value of the empty fold and absorbs masked entries; distributivity is what lets $B^{(2)} \otimes (B^{(1)} \otimes x) = (B^{(2)} B^{(1)}) \otimes x$. $\square$

Proposition 1 is the contract the language extension must enforce: **an aggregate tag is admissible iff it names a commutative monoid (for plain aggregation) or a semiring (for signed-weighted aggregation).**

3 Catalog: the admissible algebras (the aggregate tags)

1. **Additive** $(\mathbb{R}, +, \times)$ — **sum/signed_sum**. The linear convolution Bx : HSiKAN message passing $A^\pm h$, the reward $w^\top f(Bx)$, observation gather. A field semiring.
2. **Parity / Boolean** $(\{\pm 1\}, \times)$ or $(\mathbb{F}_2, \oplus, \wedge)$ — **signed_product**. The fold is the *product* of signs: Cartwright–Harary balance $\prod_{v \in e} \sigma_{ev}$. Commutative monoid under $\times$; the semiring is $\mathbb{F}_2$.
3. **Tropical / idempotent** $(\mathbb{R} \cup \{\infty\}, \min, +)$ — **tropical**. Shortest path / reachability cost; idempotent ($a \oplus a = a$). A semiring; eigenvalues are tropical.
4. **Fuzzy / residuated** $([0, 1], \perp, \top)$ with a T-conorm $\perp$ and a T-norm $\top$ — **tnorm_min/tnorm_prod**. Fuzzy rule firing. **Caveat:** a T-norm distributes over a T-conorm (a genuine semiring) only for some pairs — Gödel $(\min, \max)$ is distributive; product and Łukasiewicz are *not* fully, so they form a residuated lattice, not a semiring. The reader must therefore mark the non-distributive tags as *non-composable across nesting* (single-layer only).

4 Differential-topological structure (per algebra)

Because $B = \partial$, the chosen algebra fixes the cohomology over the *same* structure:

- **Additive:** the graph Laplacian $L = BB^\top$, Hodge decomposition $C_1 = \text{im } \delta \oplus \text{im } \partial \oplus \mathcal{H}$. Reward shaping splits into the *exact* part (gradients of a potential — Ng’s policy-invariant shaping, e.g. a telescoping progress term) and the non-exact part that reshapes the policy.
- **Parity:** cohomology over $\mathbb{F}_2$. Harary’s theorem is exactly “the sign 1-cocycle is a coboundary” $\iff$ the (hyper)graph is balanced; the convolution computes the cocycle.
- **Tropical:** the idempotent “spectral theory” is shortest-path / max-plus eigenvalues; the convolution is a Bellman operator.
- **Fuzzy:** a residuated lattice; “adjoint” is the residuum, not a transpose — no inner-product Hodge theory, which is *why* it must be treated separately.

Remark 1. “Declare the multiplication early” (Kato) is formally “declare which (co)homology / spectral theory the relation computes in.” The algebra is not a detail of execution; it is the choice of mathematical universe.

5 Spectral view

For the additive algebra the convolution is a spectral filter: with $L = BB^\top = U\Lambda U^\top$, a depth- k message-passing stack is a polynomial $p(L) = U p(\Lambda) U^\top$ acting on x . “Structure helps” iff the signal

aligns with informative eigenvectors — topological complexity (branching, cycles) is what gives L a non-trivial spectrum to exploit. For non-additive algebras the spectral object is the algebra’s own (tropical / $\mathbb{F}_2$ / lattice) spectrum; the same incidence, a different operator norm.

6 Event-based computation: walk-spikes (the tropical instantiation)

The tropical semiring $(\mathbb{R} \cup \{\infty\}, \min, +)$ is not merely “shortest path” — read the arc weights w_{ev} as **delays** and it is an *event* algebra: $\otimes = +$ accumulates delay along a walk, $\oplus = \min$ keeps the *earliest arrival*. The convolution then computes, at each vertex, the time of the **first spike** reaching it along any signed walk; a signed walk W emits a spike at time $\tau_W = \sum_{(v \rightarrow v') \in W} w_{vv'}$ carrying the walk’s sign-parity $\sigma_W = \prod \sigma$. So a vertex field becomes a *signed spike train* — “walk-spikes” — and a depth- k stack is a leaky-integrate-and-fire forward pass with structural delays. This is the same $B \otimes x$ as the additive case; only the algebra changed.

Two separable, testable claims.

1. **Representational (a bet).** For signed-link prediction the additive aggregation is *path-length-blind*: it sums all walk-parities flat. The tropical readout instead weights by the **earliest / shortest signed walk**. On a *perfectly* balanced graph the parity is order-independent and this adds nothing; but real graphs are imperfectly balanced, conflicting paths disagree, and **shorter paths are more reliable** (less sign noise accumulates). “Trust the shortest balanced walk” is a principled inductive bias the continuous code lacks — the discriminating test is tropical (shortest-signed-walk) vs. additive readout on the Bitcoin-OTC AUROC.
2. **Efficiency (near-certain).** The tropical algebra is *idempotent* ($a \oplus a = a$), so the representation is naturally **sparse and event-driven**: compute scales with spikes, not dense activations. A spiking-readout HSiKAN is the low-power, neuromorphic form of the *fast reflex* in the rate-asymmetric controller (event-based inference, the embedded/ $\sim$ kHz story), and it is the biologically apt register over the deliberative substrate.

Remark 2. *Walk-spikes are therefore not a new architecture but the tropical reading of the one convolution. With it the framework spans **rate** (additive), **spike** (tropical), **balance** (parity), and **fuzzy** (residuated) over a single signed incidence — four computational regimes, one operator.*

7 Nesting: hyperedge-on-hyperedge

A reward bundle over terms over vertices is a two-layer convolution $R = w \otimes f(B^{(1)} \otimes x)$, i.e. vertices $\xrightarrow{B^{(1)}} \text{term-nodes} \xrightarrow{w} \text{scalar}$ (the term-nodes are hypernodes; the bundle is the outer hyperedge). By Proposition 1 the layers compose when $\otimes$ distributes over $\oplus$ — so an additive bundle of additive terms is a single linear functional, but a *product* bundle of *additive* terms is genuinely two-layer and must not be flattened. The language must therefore allow a *per-edge* algebra and check compositional admissibility at the nesting boundary.

8 The extension recipe (what the implementation must do)

1. **Grammar:** an **aggregate** annotation on a hyperedge whose value names an algebra from the catalog; default **sum**. (CORE/parser — gated.)
2. **Validation:** reject a tag that is not a commutative monoid; mark non-distributive tags (product/ $\text{\Lukasiewicz}$ T-norms) as single-layer; check composition at every nesting boundary.

3. **Reader:** one `pick_semiring(tag)` returning $(\oplus, \otimes, \mathbf{0}, \mathbf{1})$; the convolution is the semiring matvec over B . Reused by reward, observation, balance, and fuzzy readers — replacing four bespoke code paths with one.
4. **Parity guard:** `signed_product` over a cycle must equal the existing `hymeko_graph::BalanceScorer`; the additive default must reproduce the current reward bit-for-bit.

One sentence. Every computation HyMeKo performs over its hypergraph is a semiring convolution $B \otimes x$; making the semiring a *declared, validated* property of each relation is the whole content of the `aggregate` extension, and it is sound exactly when each tag is a commutative monoid (semiring for signed-weighted, distributive for nesting).